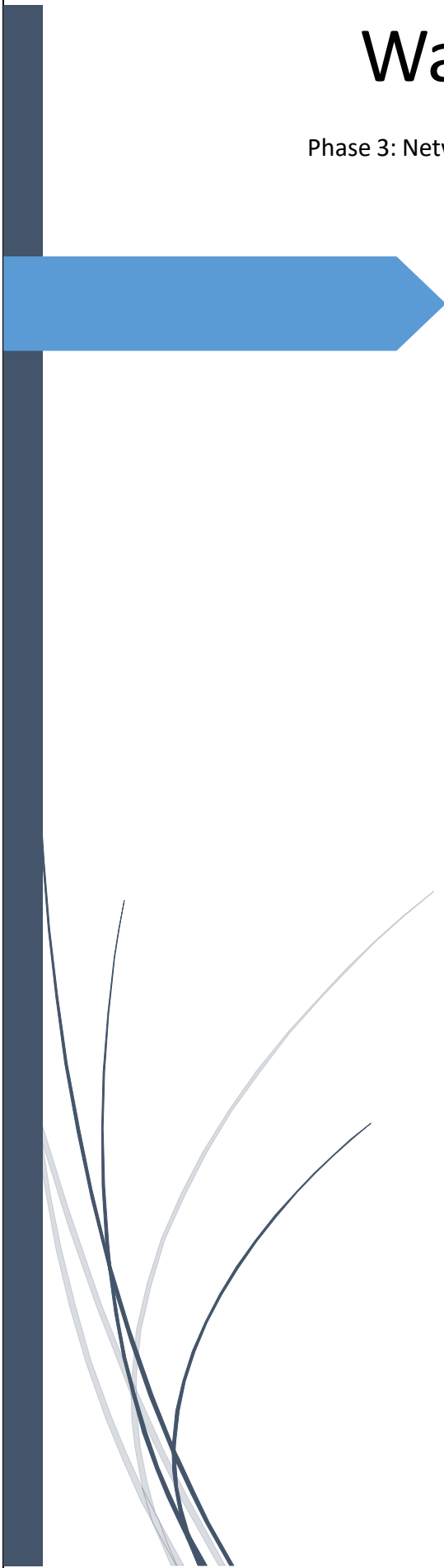




Wazuh Home Lab

Phase 3: Network Monitoring and Threat Intelligence Integration

Introduction:

This phase of the SOC home lab builds upon the previously implemented detection and response mechanisms by introducing network-based intrusion detection and threat intelligence enrichment. While earlier phases focused on host-level monitoring using Wazuh, this stage extends visibility to network traffic and external threat intelligence sources.

To achieve this, Suricata was integrated as a Network Intrusion Detection System (NIDS) to monitor and analyze network traffic, while VirusTotal was integrated with Wazuh to enrich security alerts with external reputation data. These additions simulate real-world SOC operations where both network-level visibility and threat intelligence play a critical role in identifying advanced threats.

Objective:

The objective of this phase was to enhance the SOC lab by incorporating network traffic analysis and external threat intelligence to improve detection accuracy and context.

- Network Intrusion Detection using Suricata
- Integration of Suricata logs with Wazuh
- Threat Intelligence Enrichment using VirusTotal

1) Suricata Integration (Network Intrusion Detection):

Suricata is an open-source Network Intrusion Detection System (NIDS) that monitors network traffic and detects suspicious or malicious activity based on predefined rules and signatures. It provides deep packet inspection and generates detailed logs for analysis.

1.1) Implementation:

- Installed Suricata on the Linux (Kali) machine. (**sudo add-apt-repository ppa:oisf/suricata-stable**)
- Updated Suricata rules using the built-in rule update mechanism. (**sudo apt-get update**)
- Configured the correct network interface in the Suricata configuration file.

```
# Linux high speed capture support
af-packet:
- interface: eth1
```

```
##
## Configure Suricata to load Suricata-Update managed rules.
##
default-rule-path: /etc/suricata/rules
rule-files:
- "*.rules"
```

- Started the Suricata service to begin monitoring network traffic.

```
(kali@kali)-[/etc]
$ sudo nano /etc/suricata/suricata.yaml
(kali@kali)-[/etc]
$ sudo systemctl restart suricata
(kali@kali)-[/etc]
$ sudo systemctl restart suricata
(kali@kali)-[/etc]
$ cd ..
(kali@kali)-[/]
$ sudo nano /var/ossec/etc/ossec.conf
(kali@kali)-[/]
$ sudo systemctl restart wazuh-agent
(kali@kali)-[/]
$ sudo systemctl status suricata
● suricata.service - Suricata IDS/IDP daemon
   Loaded: loaded (/usr/lib/systemd/system/suricata.service; disabled; preset: disabled)
   Active: active (running) since Sun 2026-04-12 17:58:01 EDT; 5min ago
```

1.2) Integration with Wazuh:

- Configured Wazuh agent to monitor Suricata log file (eve.json).
- Added a localfile entry in the agent configuration to forward Suricata logs to the Wazuh Manager.
- Restarted the Wazuh agent to apply changes.

```
<localfile>
  <log_format>json</log_format>
  <location>/var/log/suricata/eve.json</location>
</localfile>

</ossec_config>
```

1.3) Testing:

- Performed network scanning from the another machine using Nmap.
- Suricata successfully detected scanning activity and generated alerts.
- Alerts were forwarded to the Wazuh dashboard, where they were visualized and analyzed.

14 hits					
Mar 29, 2026 @ 12:50:59.333 - Apr 19, 2026 @ 12:50:59.333					
Export Formatted Reset view 745 available fields Columns Density 1 fields sorted Full screen					
	timestamp	agent.name	rule.description	rule.level	rule.id
	Apr 13, 2026 @ 14:53:30.950	KaliLinux	Successful sudo to ROOT executed.	3	5402
	Apr 13, 2026 @ 14:53:30.950	KaliLinux	Successful sudo to ROOT executed.	3	5402
	Apr 13, 2026 @ 14:53:09.126	KaliLinux	Suricata: Alert - SURICATA ICMPv4 un...	3	86601
	Apr 13, 2026 @ 14:53:09.126	KaliLinux	Suricata: Alert - SURICATA ICMPv4 un...	3	86601
	Apr 13, 2026 @ 14:53:09.126	KaliLinux	Suricata: Alert - SURICATA ICMPv4 un...	3	86601
	Apr 13, 2026 @ 14:53:09.126	KaliLinux	Suricata: Alert - SURICATA ICMPv4 un...	3	86601
	Apr 13, 2026 @ 14:53:08.906	KaliLinux	Suricata: Alert - SURICATA ICMPv4 un...	3	86601
	Apr 13, 2026 @ 14:53:08.906	KaliLinux	Suricata: Alert - SURICATA ICMPv4 un...	3	86601
	Apr 13, 2026 @ 14:53:08.856	KaliLinux	Suricata: Alert - SURICATA ICMPv4 un...	3	86601
	Apr 13, 2026 @ 14:53:08.856	KaliLinux	Suricata: Alert - SURICATA ICMPv4 un...	3	86601

2) VirusTotal Integration (Threat Intelligence):

VirusTotal is a threat intelligence platform that analyzes files and indicators such as hashes, IP addresses, and domains to determine their maliciousness using multiple security engines. Integrating VirusTotal with Wazuh allows automatic enrichment of alerts with external reputation data.

2.1) Implementation:

- Configured VirusTotal integration on the Wazuh Manager using an API key.
- Enabled integration for File Integrity Monitoring (syscheck) events.

```
<ossec_config>
  <integration>
    <name>virustotal</name>
    <api_key>YH28JriCCBWFiQ109H8HxEo02qtQZuCNx77NLCoLiLktuWA57drhlMq3VDjnvb4n</api_key>
    <group>syscheck</group>
    <alert_format>json</alert_format>
  </integration>
</ossec_config>
```

- Made a custom rule for virus total on wazuh manager to show alerts received from Virus Total.

```
<group name="sysmon,virustotal,">
  <!-- This rule will fire when a new .exe or .dll file is created -->
  <rule id="100100" level="7">
    <if_sid>61605</if_sid> <!-- This is the Sysmon Event ID for FileCreate -->
    <field name="win.eventdata.targetFilename">\.exe$|\.dll$|\.ps1$</field>
    <description>File - New executable or script created -> VirusTotal Scan Triggered</description>
    <group>virustotal,</group>
```

2.2) Result:

The VirusTotal integration successfully enriched Wazuh alerts with external threat intelligence. Alerts now included additional context such as detection ratio and reputation, improving the ability to analyze potential threats.

Conclusion:

In this phase, the SOC home lab was significantly enhanced by integrating both network-based detection and external threat intelligence. Suricata provided visibility into network traffic and enabled detection of suspicious activities such as port scanning, while VirusTotal added contextual intelligence to security alerts.

These additions transformed the lab from a host-based monitoring setup into a more comprehensive SOC simulation environment, combining endpoint monitoring, network analysis, and threat intelligence. This phase demonstrates how multiple security tools can be integrated to improve detection capabilities and provide deeper insight into potential security threats.